



DMAIC Project — Job Application Process Improvement (Worked Example)



This is a WORKED EXAMPLE for a fictional applicant ("Alex"). Do not present these numbers as your own. Replace every figure with your real 7-day data before using publicly.

Six Sigma Yellow Belt — DMAIC Project (Worked Example)

Belt: Yellow Belt

Method: DMAIC

Duration: 7 days

Applicant: Alex (early-career, targeting Data/Business Analyst roles)



DAY 1 — DEFINE

1.1 Problem statement

Currently, my application process takes approximately **45 minutes** per application, and roughly **67%** of applications get no follow-up because I lose track of them. This costs missed opportunities, repeated rework on resumes and cover letters, and a low response rate. Worth solving because I am actively job searching for analyst roles.

1.2 Charter

Element	Entry
Title	Job Application Process Improvement
Problem	~45 min/app; 67% with no follow-up; 53% tracked
Goal	Cut average time ~30% (45 → ≤32 min); reach 100% tracking + follow-up
In scope	Find posting → tailor → submit → log → follow up
Out of scope	Interview prep, negotiation, networking
Primary metric	Cycle time (min/app)
Secondary	Follow-up rate %, tracking rate %
Target role	Data / Business Analyst

1.3 SIPOC

Suppliers	Inputs	Process	Outputs	Customers
LinkedIn, Indeed, Handshake	Postings, base resume, base cover letter, GitHub portfolio	Find → Research → Tailor resume → Cover letter → Submit → Log → Follow up	Submitted app, tailored docs, tracked record	Alex; recruiters; hiring managers



Figure 1 — End-to-end process flow (current state)



1.4 VOC → CTQ

Need	CTQ	Metric
Apply faster	Cycle time	min/app
Never miss follow-up	Follow-up rate	%
Stay organized	Tracking rate	%
More interviews	Response rate	%



DAY 2 — MEASURE (data)

15 applications. Last 5 were **timed with a stopwatch (M)**; the earlier 10 are **recall estimates (E)** — so the baseline is **33% measured / 67% estimated** (reported honestly below).



Figure 4 — Measurement plan (what gets tracked each application)

- **Cycle time (min):** start when opening posting → stop after logging
- **M/E flag:** measured (stopwatch) vs estimated (recall)
- **Process adherence:** resume tailored, CL tailored, logged (Y/N)
- **Outcome:** follow-up sent (Y/N), response (Y/N)

#	Date	Company	Role	Source	Time	M/E	Resume tailored
---	---	---	---	---	---	---	---
1	04-02	Northwind	Data Analyst	LinkedIn	52	E	Y
2	04-04	Acme Co	BI Analyst	Indeed	38	E	Y
3	04-07	Globex	Data Analyst	LinkedIn	45	E	Y
4	04-09	Initech	Analyst I	Handshake	58	E	Y

Tracking rate	53%	100%
Response rate	13%	improve

DAY 4 — ANALYZE (map + Pareto)

4.1 Current-state waste points

Mapped 10 steps. Waste flagged at: finding the right resume version (search), writing the cover letter from a blank page (rework), re-entering identical info into each portal (duplication), and the skipped log/follow-up steps (lost info / missed opportunity).

4.2 Eight wastes (DOWNTIME)

Waste	Alex's example	Impact
Defects	Wrong company name pasted in CL twice	Med
Overproduction	Over-researching companies	Low
Waiting	Slow portal page loads	Low
Non-utilized talent	Not using templates	High
Transportation	Toggling 4 tabs/tools per app	Med
Inventory	6 stray resume versions, unclear which is current	Med
Motion	Hunting for files, reformatting	High
Extra processing	Cover letter from scratch each time	High

4.3 Pareto — avoidable minutes per application (≈30 min of the 45 is avoidable)

Time-waster	Min lost	%	Cumulative %
---	---	---	---
Cover letter from scratch (no template)	11	36.7%	36.7%
Resume tailoring (not modular)	8	26.7%	63.4%
Searching for files/right version	4	13.3%	76.7%
Re-entering form fields	3	10.0%	86.7%
Over-researching	2	6.7%	93.4%
Logging when remembered	2	6.7%	100%
Total avoidable	30	100%	



Figure 2 — Pareto (minutes lost per application; visual)

Cover letter from scratch	11
Resume tailoring	8
Searching for files/version	4
Re-entering form fields	3
Over-researching	2
Logging when remembered	2

Vital few (~77% of waste): 1) Cover letters 2) Resume tailoring 3) File searching

DAY 5 — ANALYZE (root cause)

Fishbone — “applications take too long & get lost”

- **People:** procrastination; perfectionism on cover letters
- **Process:** no standard workflow; no checklist; no follow-up trigger
- **Tools:** no tracker; scattered files; no resume/CL template
- **Materials:** each role needs different keywords; varied portals



Figure 3 — Fishbone (text sketch)

```
People -----\  
Process ----- \  
Tools -----> PROBLEM: Apps take too long + get lost  
Materials ----- /  
Environment ---/
```

5 Whys (summary)

Problem	Root cause
Cover letters slow (11 min waste)	No modular cover-letter template
67% no follow-up	No tracker with follow-up triggers
Resume tailoring slow (8 min waste)	No modular master resume

Root-cause summary

Root cause	Impact	Fix
No CL template	11 min/app	Modular CL template
No tracker	67% apps lost	Tracker + follow-up flags
No modular resume	8 min/app	Master resume w/ swappable blocks
No standard process	variability	SOP checklist
Files scattered	search time	Folder structure

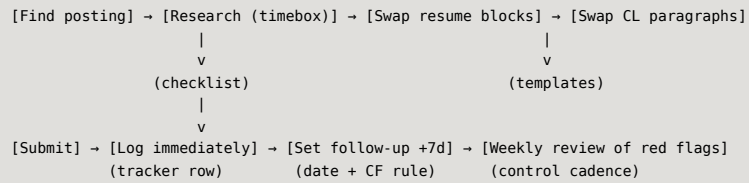


DAY 6 — IMPROVE (built)

1. **Tracker** (Google Sheets): App ID, date, company, role, source, URL, status, resume file, CL file, contact, follow-up date, follow-up sent?, response, notes — with a conditional-format rule flagging rows red when `today ≥ follow-up date` and follow-up = N.
2. **Modular master resume:** labeled skill/project blocks; tailoring is now swap-not-rewrite.
3. **Cover-letter template:** fixed intro/close + 3 swappable middle paragraphs by role type.
4. **SOP checklist:** save posting → swap resume blocks → swap CL paragraphs → submit → log → set follow-up (+7 days).
5. **Folder structure:** `00_Templates / 01_Active / 02_Submitted / 03_Interviewing / 04_Closed` + tracker.



Figure 6 — Future-state workflow (with controls built in)



DAY 7 — CONTROL (pilot + sustain)

Pilot — 3 real applications, timed

Pilot	Company	Role	Time	Checklist	Logged	Issues
---	---	---	---	---	---	---
P1	Contoso	Data Analyst	30	Y	Y	—
P2	Fabrikam	BI Analyst	26	Y	Y	portal slow
P3	Litware	Reporting Analyst	28	Y	Y	—
		Avg	28.0			

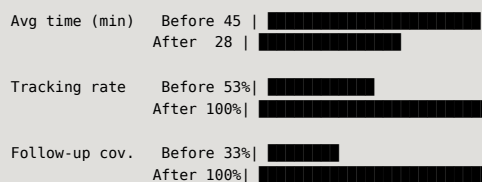
Pilot n = 3 → treat as an **early indication**, not a controlled result.

Before vs after

Metric	Before	After (pilot)	Change
---	---	---	---
Avg application time	45.0 min	28.0 min	-37.8% (≈38% faster)
Tracking rate	53%	100%	+47 pts
Follow-up coverage	33%	100%	+67 pts
Templates used	No	Yes	✓



Figure 7 — Before vs After (visual)



Math check: $(45 - 28) \div 45 = 0.378 \rightarrow 37.8\%$.

Control plan

Control	How	When	If it slips
App time	Time each app	Every app	Recheck Pareto bottleneck
Follow-up	Review tracker flags	Weekly (Sun)	Phone reminder
Tracking	Log before closing tab	Every app	Log immediately

Lessons learned

Worked	Could improve	Next
CL template saved the most time	Need 2–3 more middle paragraphs	Add role-type variants
Tracker flags ended missed follow-ups	Want mobile access	Move to Google Sheets app
Checklist kept steps consistent	Slightly long	Trim to 5 steps

Results summary (this example)

PROJECT: Job Application Process Improvement (Yellow Belt, DMAIC)
 TYPE: self-directed practice project | DURATION: 7 days

BUILT: modular resume + cover-letter system · tracker with follow-up flags · SOP · folder structure

RESULT (directional; self-directed, pilot n=3):

- Avg application time ~38% lower (45 → 28 min)
- Tracking 53% → 100% · Follow-up coverage 33% → 100%
- Baseline: 33% measured / 67% estimated

TOOLS: SIPOC, VOC/CTQ, process mapping, 8 wastes, Pareto, fishbone, 5 Whys, control plan



When you run your own 7-day project, duplicate this page and replace every figure with your real data.